Estimated Dynamic Equilibrium Model:

Supply and Demand

as a Sample Path of a Stochastic Process

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SUMMARY

To this day Efficient Market Theory remains the only accepted theory explaining mechanics of price formation. Although it has an extraordinary body of evidence, it fails to explain the most important market anomalies such as bubbles and crashes. Numerous rival theories developed to explain the anomalies remain controversial as nosy historical data contains evidence to all of them; even when they are mutually exclusive. To avoid data mining for another controversial theory, I rely on realistic assumptions and math to build an agent-based model of price formation mechanics. As the proposed model not only explains the formation of market anomalies but also provides an explanation of why real world data supports mutually exclusive hypotheses, it may become the single uncontroversial theory for price formation mechanics.

Develop and maintain techincal solutions to advance company's growth efforts.

- **Refine and improve internal operations and tools**

- **Data aggregation and collection from variety of sources including field data**
- **Use aggregated data and trending analysis for bi-weekly and monthly production estimates**
- **Maintain and analyze databases to calculate weekly KPIs**
- **Assist Business Development and Executive initiatives as needed**

SPECIFIC AIMS

The goal of this research is to present a fresh view on Supply and Demand and examine the connection between the divergence of opinion and market behavior in a controlled environment. The empirical researches have failed to reach a consensus on the effects of divergence of opinion. To avoid the complexity and noise of real-world data, I attempt to use agent-based modeling to stage experiments, where a change in the input parameter can be isolated from the noise, and thus may be interpreted. The model shows that in the presence of divergence of opinion, supply and demand become a sample path of a stochastic process, that may never stumble upon

equilibrium. I test the model against well know real-world scenarios to prove that it may be accepted as an initial framework to model price formation mechanism.

BACKGROUND AND SIGNIFICANCE

"The very concept of uncertainty implies
that reasonable men may differ in their forecasts."
Edward M. Miller, 1977

Centuries of empirical research failed to provide a single model to explain the mechanism of price formation. (Motty, 2017) Recent developments in data mining successfully provide historical evidence to mutually exclusive

hypotheses. (Kim, 2006) The divergence of opinion premium and discount hypotheses have been proved and disproved numerous times over past 50 years. Abundant and noisy financial data helps find evidence for any wild theory, so I revert to pure logic and math modeling to present an elegant and concise framework to explain the most crucial market anomalies, such as bubbles and crashes. My model is a dynamic extension of Edward M Miller view on price formation mechanism.

In 1977 Edward M Miller presented a simple model that explained how a minority of over-optimistic investors define a market price that is above mean valuation of the potential investors. "Suppose that any single investor is able to purchase only one share (perhaps because of limited funds) and there are N shares available. It is clear that the shares will end up being owned by the N investors with the highest evaluation of the return.

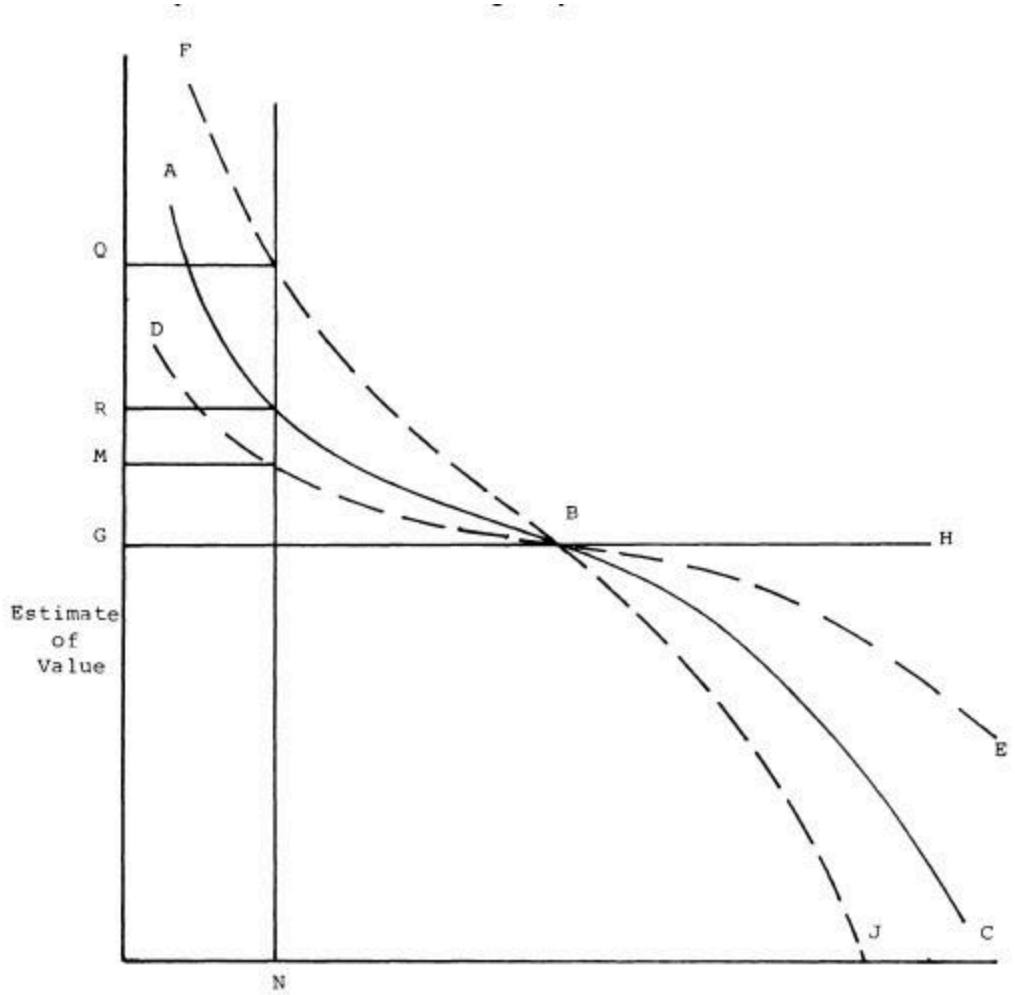


FIGURE 1.

From curve ABC it can be seen that N investors estimate the final value to be R or above. The selling price of the stock will be R." Curve FBJ represents a demand curve with the greater divergence of opinion, which leads to a higher value Q. Similarly, when the divergence of opinion is small, curve DBE intersects demand curve N at the value M. The high divergence of opinion leads to a severe overvaluation that results in lower future returns. A small divergence of opinion results in higher future returns, which is known as

Miller's (1977) overvaluation hypothesis or “divergence of opinion premium hypothesis” (Chansog Kim, 2006).

To provide empirical evidence to his hypothesis, Miller examines prices for new stock issuance. “If the divergence of opinion about a stock changes (leaving the average evaluation unchanged) the market price should change. For instance, if risky stocks become less risky over time, their prices should drop. There is some reason to believe that the stocks about which there is the greatest divergence of opinion at any one time tend to have this divergence of opinion narrow over time. This is primarily because the passage of time often resolves certain uncertainties about the future of a company. When this happens, the market valuation will tend to decline relative to the securities whose yield is more easily estimated. This provides an additional reason for the returns on the riskiest securities to be lower than on less risky Securities. Although new issues frequently show a sharp spurt in price immediately following issuance due to underpricing by the underwriter (McDonald and Fisher, Reilly and Hatfield, Friend and Longstreet), this is followed by smaller price appreciation than "seasoned" stocks show when measured over a period of one to five years. (Stigler; Shaw; Shaw and Herman; McDonald and Fisher; Friend and Longstreet)”

Miller's narrative is so elegant and intuitive that it cannot be ignored. As Steven Moffit has put it, "Although Miller's model has largely been ignored, it has lost none of its relevance." (Moffit, 2017) Chansog Kim provides a careful overview of the heated empiricist battle between supporters of divergence of opinion premium and discount hypotheses. The discount hypothesis states that divergence of opinion proxies risk, so the higher the divergence of the opinion the riskier is the asset and the higher the discount. One consequence of this argument is that "the greater the disagreement among investors about the value of a stock, the lower its market price relative to its true value and, therefore, the higher its future return." (see Williams (1977), Mayshar (1983), Epstein and Wang (1994), Merton (1987), and Varian (1985))."

The premium hypothesis has as many supporters. Chansog Kim points out that many researchers provide evidence in favor of a positive association between future stock returns and dispersion in analysts' earnings forecasts. (see Cragg and Malkiel (1968), (1982), Friend, Westerfield, and Granito (1978), and Harris (1986). His research, "Divergence of Opinion and Equity Returns" provides evidence of significant association between future stock returns and divergence of opinion among analysts, which he interprets to be consistent with the divergence of opinion discount hypothesis.

Steven D. Motty sums up the results of this relentless debate:

> *"Though we have over two centuries of financial market history and various theories of price formation, today there exists no single theory acceptable to market practitioners, yet rigorous enough to satisfy market theorists. The only major theory that purports universality is the efficient market theory, which fails to explain some of the most important market phenomena, e.g. bubbles and crashes."*

I use Edward Miller argument to build an accurate dynamic model of price formation. I attempt to avoid unrealistic assumptions and restrict model to the essential variables so that results can be interpreted and generalized to various markets. The model may be used to explain different economic phenomena: double exponential bitcoin price behavior, silicon valley real estate market failure to reach equilibrium, art prices, as well as 'sticky' behavior of contractors on construction market and high bankruptcy levels of small construction companies. As the model fits well with a random walk and winner's curse theories, the model can be further cross-validated with other theories to find its boundaries.

The model warns that machine learning predictive algorithms may amplify human errors and serve as effective catalysts for market destabilization. If they are built on an over-optimistic or over pessimistic market data, they

effectively exaggerate human estimation errors. Consider an appraisal algorithm for real estate market. Its goal would be not to estimate the current home value, that may be already very well overpriced but estimate the value of the winning bid, which is the price with the maximum positive valuation error. Thus overvaluation becomes a self-fulfilling prediction. If an algorithm feeds upon human overvaluation, it promotes human overvaluation.

At the same time, if given the right information, the same algorithm can diminish the error and help to stabilize the markets thus preventing the random walks. Moreover, it may help balance supply and demand by providing early notice of market shortage/surplus and predicting adequate magnitude of response. This improves market elasticity and helps it to reach equilibrium faster and stay there for a prolonged amount of time. Today, economists shall finally find the answer to the mechanics of price formation, otherwise, we can expect catastrophic bubbles and crashes. Just as we have seen with Dotcoms, CDOs, and ICOs.

PRELIMINARY DATA

The model presents market price as a sample path of a stochastic process that may never stumble upon equilibrium. The classic view of these functions is $Q_s(p)$, $Q_d(p)$, where quantity supplied $Q_s(p)$ and quantity demanded $Q_d(p)$ are functions of the market price, p. A market equilibrium is the state when supply and demand forces are balanced and occurs when quantity supplied equal to quantity demanded:

$$Q_s(p_{eq}) = Q_d(p_{eq})$$

In the absence of external influences, the market clearing price, p_{eq} and the quantity stay constant. As we are interested in the mechanics of price formation let us add time variable to the equation:

$$Q_s(t)(p(t)) = Q_d(t)(p(t))$$

Then market equilibrium state is achieved when all the variables are constant:

$$(\Delta p / \Delta t) = \left(\Delta Q_s / \Delta t\right) = \left(\Delta Q_d / \Delta t\right) = 0$$

According to the classic view this state is inevitable, our model will show that there are certain conditions for stable market and even then the variables tend to hover around the equilibrium rather than being constant.

Let us define market clearing price at time t as outcome of market clearing algorithm, A, which depends on quality supplied, Qs(t), quality demanded, Qd(t), and population valuation error, ε:

$$p(t) = A(Q_s(t), Q_d(t), \varepsilon)$$

where quantity demanded and quantity supplied depend on the price change:

$$Qs\ (t) = Qs(t-1) + Cb * sgn\left(\frac{dp}{dt}(t-1)\right)$$

$$Qd\ (t) = Qd(t-1) - Cs * sgn\left(\frac{dp}{dt}(t-1)\right)$$

The market price, quantity supplied and quantity demanded, p, Q_s, Q_d, are stochastic functions that depend on the error term ε, , balancing parameters Cb, Cs , time, t and each other. This dissection allows our model to present Supply and Demand as a realization or a sample path of a stochastic process. Preliminary data shows that such view of supply and demand can hint mechanics behind market anomalies, but also shows how a change in parameters shifts the distribution of random path destinations.

To make preliminary results interpretable, the initial model has only a few parameters. Quantity demanded equals the number of buyers on the market. If the market price falls, Cb , the balancing parameter determines how many additional buyers enter the market. The presented simplistic

model assumes Cb to be constant; this helps to focus on the more important parameter that represents the divergence of opinion.

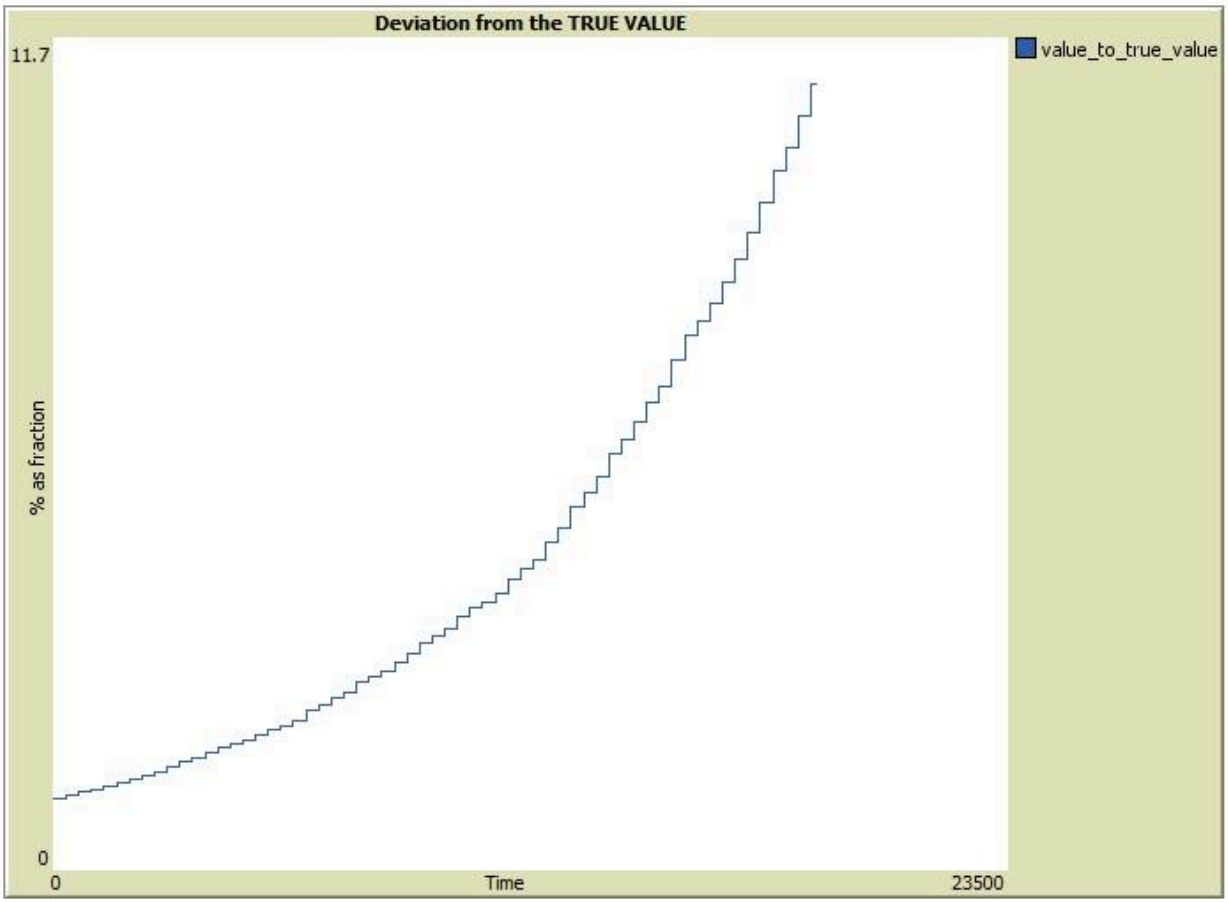
```
;; this procedures sets up the model
to setup
  clear-all
  ask patches [ ;; give value to the patches, color it shades of green
    set cycle init_patience - 1
    set true_value ( random-float 10.0 * 100 + 500 )
    set value true_value
    set pcolor green ;; change the world green
  ]

  create-sellers number-of-sellers [  ;; create the initial sellers
    setxy random-xcor random-ycor
    set color red
    set shape "house" ;; look up the smile face
    set size 2 ;; increase their size so they are a little easier to see
    set epsilon random-float init_epsilon  ;; set the individual errors range for home value estimation
    set patience init_patience
    set number_of_bids 0
  ]

    create-buyers number-of-buyers [  ;; create the initial buyers
    setxy random-xcor random-ycor
    set color blue
    set patience init_patience
    set lowest_bid_to_value 2
    set shape "person" ;; look up the smile face
    set size 2 ;; increase their size so they are a little easier to see
    set epsilon random-float init_epsilon  ;; set the individual errors range for home value estimation
  ]

  set response_time init_patience - 1
  reset-ticks
end
```

To present divergence of opinion, model equips every agent on the market with a unique stochastic estimation function:

$$E_{home}^{agent}\ (t)\ = p_{home}(t)\ +\ error\ (\varepsilon^{agent}, t)$$

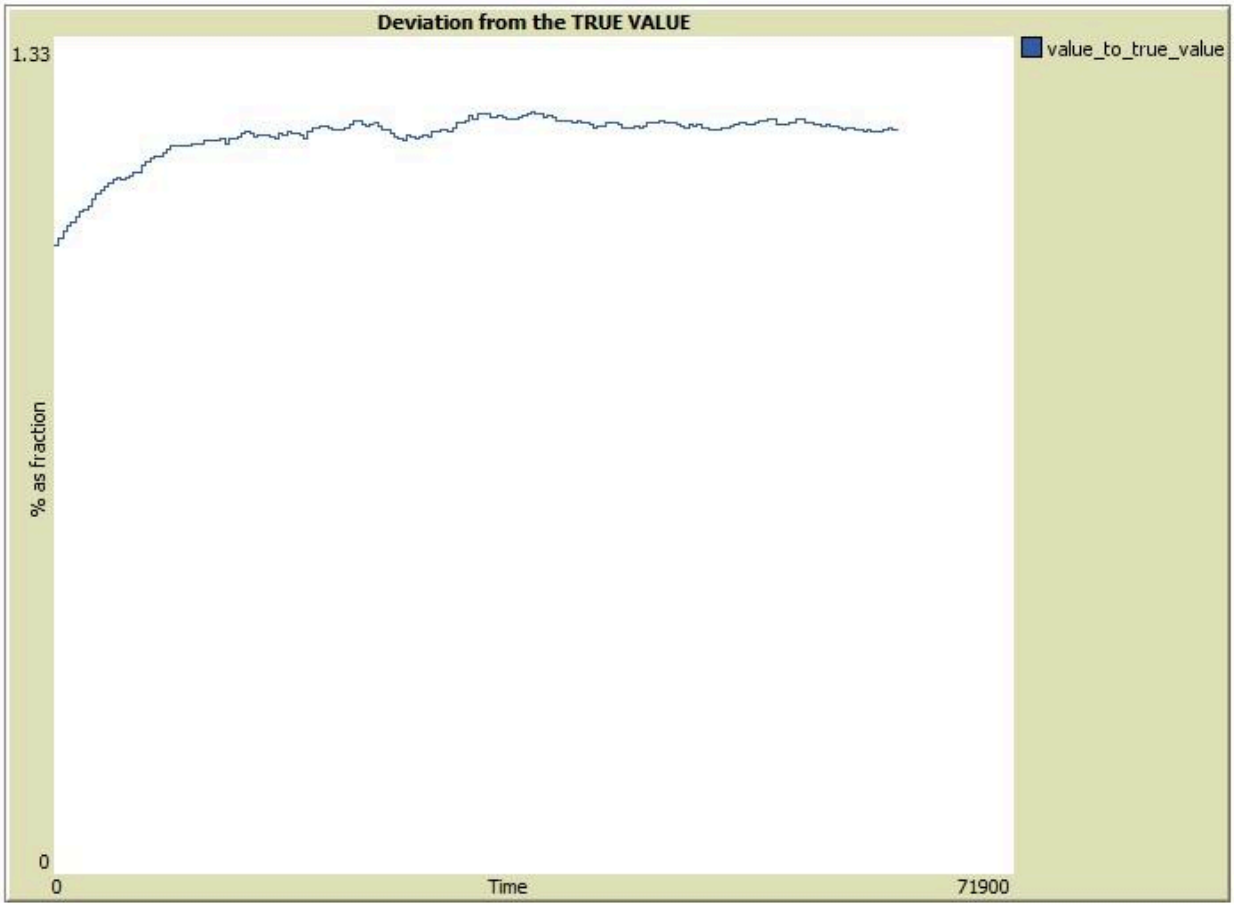
The fair price, $p_{home}(t)$, is the value of the asset given by the most accurate and precise estimation function, which is unknown to the agents. Agents have less precise estimation functions, that although centered around the fair price, give estimations with a random error. The random error come

from uniform distribution $\left(-\varepsilon^{agent}, \varepsilon^{agent}\right)$, where $\varepsilon^{agent} \in (0, \epsilon)$ is unique for every agent. So while some agents are able to give estimations close to the fair price, others estimates vary in a wide range.

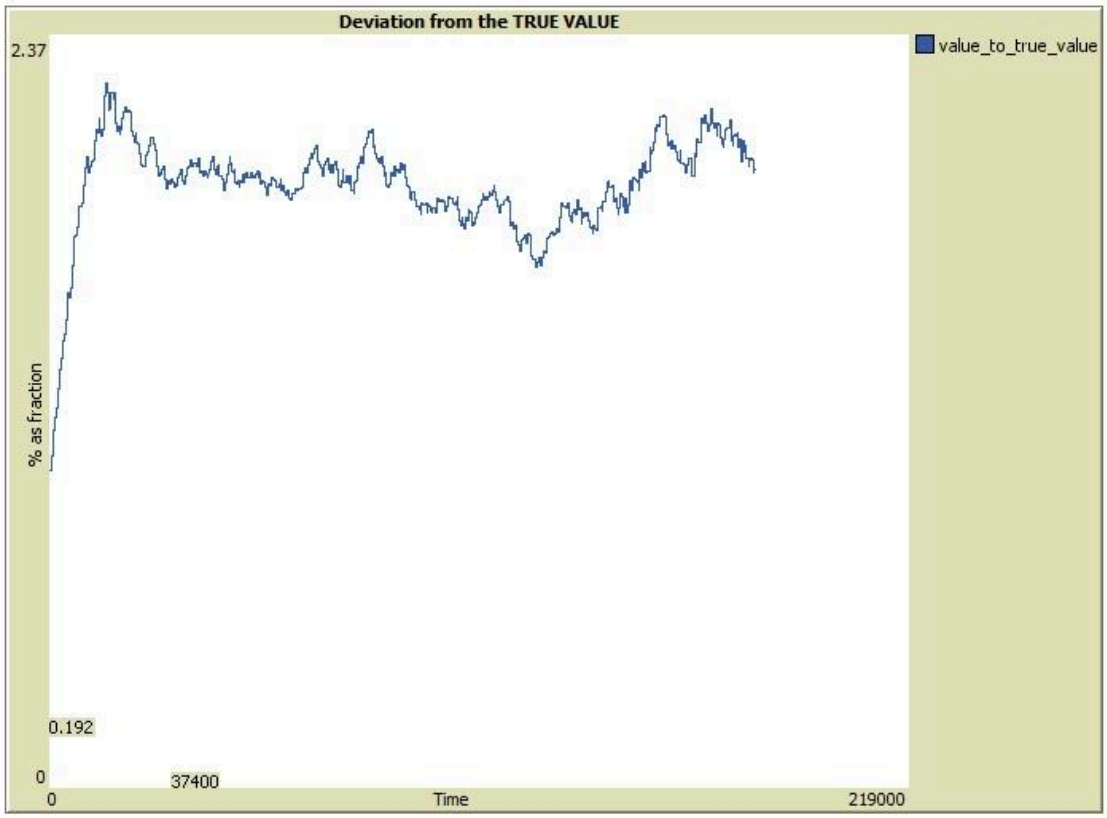


The model shows that behavior of the market depends on the balance between the balancing parameter, and v, the parameter defining the magnitude of estimation errors. For instance when Cb = 0, the market fails to adjust the number of sellers and buyers on the market. Under this condition, overbidding buyers cause exponentially increasing market price, even in the absence of any shortage, when the number of seller equals the

number of buyers. Failure of the increasing prices to provoke change in the ratio of buyer and sellers leads to the market bubble. To make things worse, one can make the balancing parameter negative. Then the higher price goes up the more buyers and the fewer sellers are on the market. (Now that looks like Silicon Valley Real Estate market.) If one also increases the estimation error terms, while prices are on the rise, this results in impressive double exponential price growth. This is exactly what was observed on bitcoin markets.



If the balance power is just right to offset errors in valuations, then after initial charge(fall), the market reaches a plateau and starts to hover around an equilibrium ratio of buyers and sellers. However, if error term is a bit higher than the balancer can handle, the market starts a “random walk on a leash”behavior. It looks just like a little girl trying to walk a humongous dog. She tries to walk the dog in the right direction, but as the dog is stronger, they may end up anywhere in the neighborhood. The balancer still prevents the exponential growth but does not have enough power to offset the pull of the random walk. Hence the price level goes randomly up and down, imitating business cycles. A powerful bull run can drag prices up to a new equilibrium price level, while bear markets can drag prices way down. With enough randomness and patience, I have observed prolonged bull markets as well as sudden catastrophic nosedives.



The simulations capture the connection of divergence of opinion and market returns that divergence of opinion discount and premium hypotheses fail to notice. Unlike noisy real-world data, the model isolates and interprets the impacts of the divergence of opinion. A significant divergence of opinion may result in exponential market growth and thus high returns, but also may lead to a random walk which results in random returns. It also can lead to stable returns if the number of buyers and sellers is rebalanced to offset the pulls of over/underestimation.

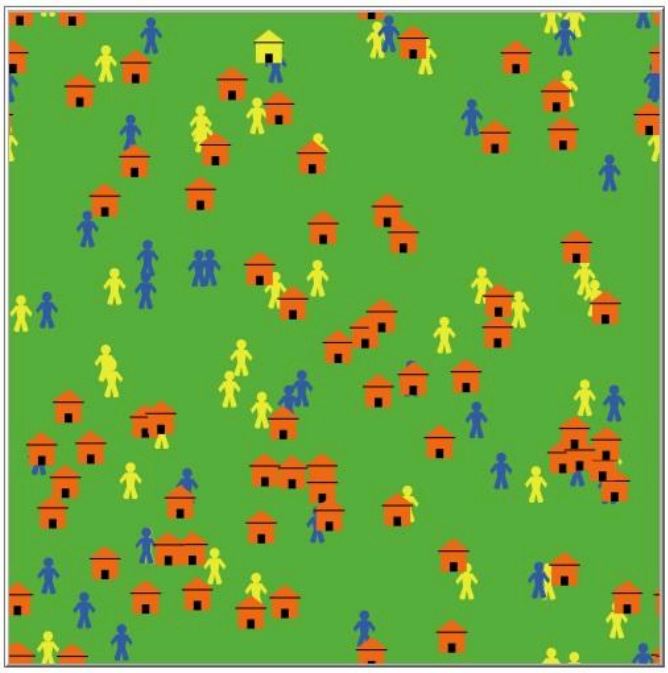
To find empirical evidence of the mechanics behind the proposed model, a curious researcher could examine assets that present on one market and have similar underlying value, but have different estimation complexity and see if the market price for easy to estimate products shows a more stable behavior when treated for the common trend. A curious extension of the model may simulate the effect of shocks on a thin market. Imagine a growing market that has a diminishing number of sellers and buyers such that the market price is defined by a few sales. Then the market is shocked with a large number of sellers, the outcomes of such an event are hard to calculate but may be evaluated with the proposed model.

MODEL DESIGN

The proposed agent-based model, called Estimated Dynamic Equilibrium Model (EDEM), describes the connection between the divergence of opinion and the market price. As the model should generalize and be interpretable, it uses a few simple variables and as few realistic assumptions.

One assumption is that every home has a fair price. The fair price is a value given by the most accurate and precise estimation function. This function is

so complex and demands so many inputs that agents are unable to figure it out. However, the agents act rationally and use all available information and best estimation functions to define a home price. The functions and the available information are not the same for every agent, which results in the divergence of opinion. Every agent predicts a home price with a random error given by assigned unique random function. The error is random as otherwise a better estimation function would have been constructed by the agent, which violates the assumption that people are smart enough to use the best possible estimation functions. The final assumption states that a seller sells at the highest possible price, a buyer buys at the lowest possible price and the prices of deals are available to both.



The model is staged in a square world, where every patch is a home. According to the true value assumption, every home is assigned a fair price. There is no restriction on the distribution of the fair prices. The distribution is irrelevant because the fair prices are used simply to measure the deviation of a market value from the true value. The deviation is simply:

sum(market_value) - sum(true_value), for all homes (patches)

Our square world is populated by people who want to sell or buy a home. According to the second assumption, each buyer and seller has an estimation function that estimates the true value with a random error. To satisfy the second assumption, we equip every buyer and seller with an estimation function with normally distributed error:

True_value * (1 +- random(variance))

The variance (variable #2) of the error is specific for every agent, so some have almost perfect estimation functions with small error term, while others have large errors that may result in substantial over- or under-bidding.

```
to sell
  if breed = sellers [  ;; seller?
    ifelse patience = init_patience [
      if any? sellers-here [
        setxy random-xcor random-ycor
      ]
      set price ( value + (value * ((random-float epsilon * 2) - epsilon) / 100 ))
      ;;set bid price
      set patience (patience - 1)
    ]
    [ifelse patience = 0
      [setxy random-xcor random-ycor
      set patience init_patience
      set bid 0
      set total_bids 0
      set number_of_bids 0
      set avg_bid_to_value 0
      set best_bid_to_value 0
      set color red
      ]
      [set patience patience - 1 ;; if patience not over, keep selling
        if bid < price [
          set price price * 0.9999
          ;;set bid price
        ]
      ]
    ]
  ]
end
```

Although buyers and sellers have similar estimation functions, they behave differently. Sellers are placed in a home and cannot move until they sell it. They start to sell at a price given by their estimation functions. If the asking price is so high that no sufficiently high bids are received, then the seller may lower the price. The sale is made when seller runs out of patience. The price of the sale becomes available to all agents on the market. If no bid is above the asked price, no sale is made.

```
;; buyers procedure, the agent changes its heading
to wiggle
  ;; turn right then left, so the average is straight ahead
  if breed = buyers[
    rt random 90
    lt random 90
  ]
end

;; buyers procedure, the agent moves which costs it energy
to move
  if breed = buyers[
    forward 1
  ]
end
```

Buyers can move. As soon as they randomly land on the square world, they start to wigle and step forward. If they hit a home for sale, they bid a price generated by their individual estimation function and move on. There is no explicit limit to the number of bids. Buyers make as many bids as many homes they meet. As the process is stochastic, some buyers may place tens of bids, while others may end up just having a good walk without stumbling upon a single home. When their patience runs out buyers buy a home at the least possible price, which has to be the highest proposed price, so that seller accepts it.

The patience is an important variable that indirectly defines how many bids buyers submit and sellers receive. If sellers receive several bids with random error, the chances are they will be able to sell the home at a much higher

than market value price. When all homes sold at a higher price; this becomes a trend so the market value of all homes in the neighborhood (or our square world) are adjusted. Unsurprisingly, this continuous adjustment results in an exponential price growth even when the number of sellers and buyers are equal.

But markets are not rigged to be increasing. On a falling market, when buyers chose among blocks of homes for sale, they have a chance to buy a home substantially below the market. If buyers walk around long enough to make a few winning bids with negative estimation error. This leads to decrease in the market valuation of the neighborhood, thus next round starts at a lower price level. The market keeps falling until the balance in the number of buyers and sellers is not restored. A balancer function determines if the balance ever reached.

Real World markets have adjustment mechanisms to slow down the growth (alas they are not working properly and not all the time). Therefore, I add a simple supply-demand balancer to simulate the market adjustment mechanism. If the price has increased in the previous round, the model adds a seller as more people should be willing to sell their home at a higher price.

```
to balance
  set response_time response_time - 1
  if response_time = 0 [
    set response_time init_patience
    ifelse (count buyers with [color = yellow] != 0) and ((sum [ lowest_bid_to_value ] of buyers with [color = yellow]) / (count buyers with [color = yellow])) < 1 [
      ask n-of 1 buyers [ hatch 1 [ lt 45 fd 1 ]]
      ask n-of 1 sellers [die]
     ]
    [ask n-of 1 sellers [ hatch 1 [ lt 45 fd 1 ]]
       ask n-of 1 buyers [die]
     ]
  ]
end
```

At the same time, fewer buyers can afford to pay the steeper price, so one buyer is forced to leave the market. Similarly, when real estate prices decrease sellers are less willing to sell, while more buyers can afford the purchase. If price decreases a random buyer hatched while a random seller dies. Even such a simple balancer is able to kill the exponential price growth unless the average error is too large.

As the model has only the essential parameters, it can be used as a framework for extensions. For instance, the sellers can be converted to realtors, who can earn commission, which is a good way to test realtor's strategies that may lead to success. Or blind auctions can be changed to auctions where buyers may see other bids and correct the proposed price. The model also can be converted to reflect construction market, where buyers become contractors and thus the smallest bid wins; but if a contractor loses more than a certain amount of money he leaves the market. Moreover, the larger contractors may be allowed to participate in many sales while small contractors can sign only one contract at a time.

Additional research is needed to prove that the model is accurate in simulating various market phenomenon encountered in a variety of markets. If it is, then a solution to the prolonged debate over divergence of opinion finally may be found. I believe this simple model may be the right direction to build much needed "single theory acceptable to market practitioners, yet rigorous enough to satisfy market theorists." (Motty 2017)

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